

Merge Guide

NHAA Trauma Triage System -- Final Assembly, After All Five Phases Are Done

When to Use This Document

Only after all five people have finished their own Phase 5 packaging step independently. This is not a build phase for any one person -- it is the one step where the five pieces come together.

The Merge Order

1. Start with **Person 2's** `SafetyGateModule` and **Person 1's** `nhaa_core_pipeline.py` -- these two must agree on the shared contract exactly, since Person 1's output is Person 2's input. Run Person 1's `process_audio_chunk` output directly into Person 2's `process_bucket_change` function and confirm no shape mismatch errors occur. If there is a mismatch, fix it by adjusting whichever side deviated from the originally agreed contract -- do not change the contract itself at this stage.
2. Add **Person 2's** confidence-gating, scheduled-checkin, and supervisor-queue functions (`confidence_gate.py`, `safety_checkins.py`, `supervisor_queue.py`) into the same flow: call `should_force_human_review` immediately after Person 1's pipeline produces `overall_confidence`; call `schedule_safety_checkins` immediately after any Low or Moderate bucket is reached; call `flag_for_supervisor_review` immediately after any bucket is reached, regardless of level.
3. Add **Person 4's** `nhaa_wow_factors.py` functions into the same pipeline: call `detect_duress_code` on the incoming audio's DTMF stream alongside Person 1's ASR call, and call `find_and_book_slot` (passing the case's risk-vector summary as `case_summary`) whenever Person 2's gate produces a `counseling_referral` or `legal_aid` action in the executed state.
4. Connect **Person 3's** frontend to the now-combined backend by replacing their mock-data WebSocket URL with the real one. Test every screen against real data flowing through for the first time -- expect to find and fix several small shape mismatches here, since this is the first time all pieces have run together.
5. Add **Person 5's** `nhaa_frontier.py` module last, since it depends on the fully working core pipeline and safety gate already being in place -- wire `check_pattern_correlation` to run on every new Critical/High case, and confirm `prepare_asha_handoff` correctly refuses to run on any case that hasn't passed Person 2's real senior-approval gate.

6. Run one full end-to-end test, additionally confirming: a low-confidence case is correctly forced into mandatory human review, a Low-bucket case correctly schedules 72-hour/7-day checkins, and every Critical case correctly appears in the supervisor review queue: a real recorded audio clip, played through the complete merged system, watched on Person 3's real UI, checking that the SVI climbs, the correct actions appear, the senior-approval gate holds, and the audit log's hash chain verifies correctly at the end.

If Something Doesn't Merge Cleanly

The most likely failure point is a small shape mismatch between what one person's module actually outputs and what the shared contract specified. Fix these by referring back to each person's individual PDF's "Shared Contract" section -- that section is the source of truth all five were building against, so the fix is almost always making the mismatched piece match that spec, not changing the spec itself this late in the process.